

inMediaVeritas

A Decentralized News Verification Platform Using zk-SNARKs and Blockchain

Jakub Ivácek, Patrik Baran, Tomáš Tisovský, Vojtech Babinský, Tomáš Miština, Matúš Hrkeľ, Richard Gazdík, Ivan Homoliak

Slovak University of Technology, Faculty of Informatics and Information Technologies
Brno University of Technology, Faculty of Information Technology

BRAINS 2026

Motivation

Problem

- Misinformation spreads faster than it can be checked
- Fact-checking relies on editors, platforms or a designated verifier
- Automated tools (ClaimBuster, LLMs) give no verifiable provenance

Gap

- Hashing a file proves it was not changed, not that the event happened

Goal

- Auditable community attestation of real-world events
- No single party decides what is true





Proposed Solution: inMediaVeritas DApp

Contributions

- End-to-end architecture: smart contracts, IPFS, decentralized identity, zk-SNARKs
- Credibility-weighted witness workflow with quorum and slashing
- Working proof of concept with mobile proving and on-chain gas measurements

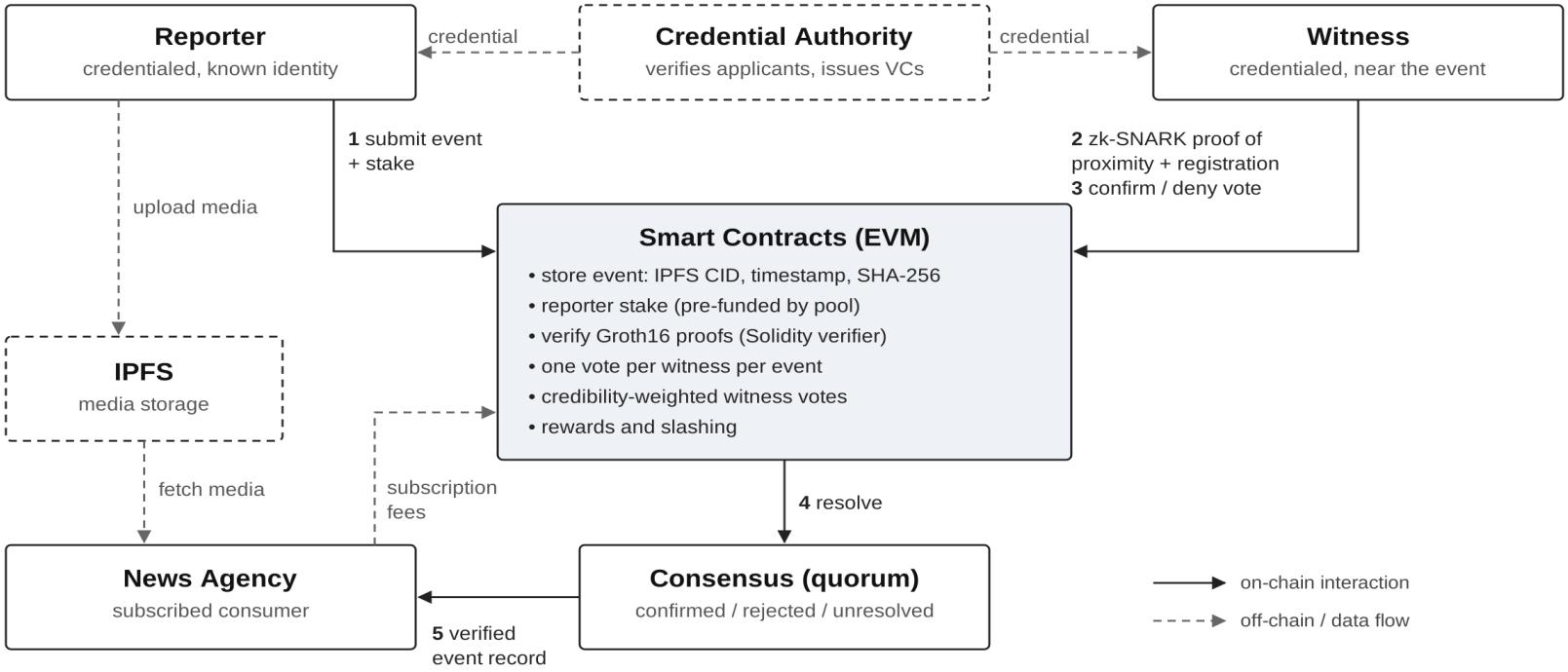
Technology stack

- Ethereum-compatible smart contracts (Solidity)
- Circom circuits, Groth16 zk-SNARKs
- IPFS media storage, DIDs + verifiable credentials



inMediaVeritas

System Architecture



Reporter Layer (journalists)

A submission commits on-chain

- IPFS content identifier (CID) of the media
- Timestamp
- SHA-256 integrity hash

Integrity

- Any change after submission breaks the hash and the CID

Accountability

- Admitted only after identity verification
- Reputation stake on each report; gas pre-funded by the protocol



Witness Layer (community)

Credentialed users near the event

- Vote confirm or deny
- zk-SNARK proves: inside the event radius and a registered participant
- Exact location and identity stay private

Resolution

- Votes weighted by credibility (past agreement with consensus)
- Configurable minimum quorum

Confirmed

quorum reached, weighted majority confirms

Rejected

quorum reached, weighted majority denies

Unresolved

no quorum in time; reporter stake returned

News Agencies and Incentives

Agencies subscribe to access

- Resolved events and their media
- Witness votes and authenticity proofs

Fees fund the protocol pool

- Reporters: gas reimbursement + reward after resolution
- Witnesses: share by contribution to consensus
- Dishonest stakes slashed back to the pool
- No personal capital at risk for participants



Identity and Cryptography

Identity

- Mobile wallet holds credentials (DIDs + VCs), generates proofs
- Keys in Secure Enclave / Keystore, unlocked biometrically
- Authority app verifies applicants, issues credentials, keeps an audit log
- Authority handles onboarding only, never consensus



Location circuit

- Circom, Groth16 over BN254 (Baby JubJub inside the circuit)
- 192-byte proof, checked by a generated Solidity verifier

Evaluation

< 300 ms
proof generation on phones

192 B
Groth16 proof size

~291–304k
gas per zk-enabled vote

450,615
gas to create an event (1 media)

Total proving time (ms)	Identity	Location
Pixel 6 Pro (native)	276.66	254.31
Samsung S23 Ultra (native)	186.53	205.63
Desktop, SnarkJS cold run	2,575	2,362

- Proof verification is ~74–75% of a zk vote's gas
- ≈ 0.0003 ETH (~$0.80) per zk vote at 1 gwei, a recent mainnet average
- Fees are covered by the protocol pool, not by users
- Measured speed and cost, not resistance to cheating (future work)

Mobile: 10 runs per device. Gas: Remix JS VM, Solidity 0.8.13, optimizer on.

Limitations and Next Steps

Witness coverage

Rural areas, odd hours or fast-moving events may not reach quorum

Credential authority

A captured authority could block users or admit Sybil identities

Collusion

No bound yet on colluding witnesses; incentives need game-theoretic analysis

Not deepfake detection

Media faked before submission keeps a consistent hash

Next: adversarial and economic analysis, L2 deployment, distributed pinning (Filecoin / Arweave), more decentralized credential issuance

Conclusion

inMediaVeritas shows it is feasible to combine

- Mobile zk-SNARK proofs in under 300 ms
- Blockchain-attested witness consensus at a measurable gas cost
- Decentralized identity and IPFS media integrity

A systems proof of concept

- An extra verification layer for newsrooms
- Not a replacement for editorial judgment



inMediaVeritas

Live Demo

Image sources

- The New York Times, 2 Apr 2024 (Damascus strike)
- The New York Times, 21 Dec 2022 (Ukraine war)
- Photojournalists on War, “Risking it all: the life of a military journalist”
- Designboyo, “The 10 best media brand logos”